

Hrvoje Jerković, PhD

Data Architect | Principal Data Engineer

Šibenik, Croatia | Remote B2B | European working hours

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Online CV | LinkedIn

Profile

Data and solution architect combining hands-on engineering, technical leadership and education. Data and ML project experience since 2017, with a broader background in information systems. Specialises in lakehouse modernisation, lineage, reliable pipelines and AI-assisted migration across financial services, UK consulting and the public sector.

Core expertise

- **Architecture and governance:** lakehouse/DWH, Medallion, relational and dimensional modelling, metadata, table/column lineage, data quality, access controls and auditability.
- **Engineering:** Python, SQL, dbt, Spark, Delta Lake, Airflow, Kafka, Snowflake, PostgreSQL and Oracle; batch, streaming, APIs and CDC patterns.
- **Cloud and delivery:** AWS production projects; GCP metadata/CDC prototyping; Azure Databricks workshop and lab development; Docker, Git and CI/CD practices.
- **Consulting and AI:** requirements discovery, architecture trade-offs, technical coordination, mentoring and workshops; AI-assisted ETL analysis and validated migration workflows.

Selected experience

Data Architect / Principal Data Engineer

Raiffeisenbank Austria d.d., Croatia (RBA) | Sep 2025 - Feb 2026 | Project case study

- Designed and implemented a lakehouse migration framework for an Oracle/PostgreSQL estate of approximately 4,300 ETL mappings and 1,800 landing tables. Defined Bronze/Silver/Gold layers and reusable dbt models targeting Spark and Delta Lake.
- Developed automated reverse engineering of ODI, DataStage and PL/SQL flows, reconstructing table- and column-level lineage across more than 20 source systems. Produced dependency graphs, SQL traceability and structured metadata for downstream automation.
- Built orchestration with nine specialised AI agents to translate legacy logic into dbt SQL. Added dependency ordering, SQL compilation/execution, checks against production reference outputs and failure escalation.
- Simplified the migration pattern by removing a separate materialised data-quality layer, eliminating approximately 340 materialisations. Handed over reusable models and pipelines to the delivery team.

Principal Data Engineer / Data Consultant

Intuita Consulting, London, UK | Aug 2022 - Feb 2024

- VIOOH / JCDecaux: led SQL/dbt model upgrades and maintenance in a Kafka-to-S3-to-Airflow-to-Snowflake pipeline handling approximately 1-5 million nested JSON records per hour. Investigated data issues using DataDog and Grafana.
- Designed and implemented a pipeline anomaly-detection and notification application using Python, SQL, AWS Athena, Lambda, Glue and Snowflake integrations.
- Altus Group: led data modelling and integration of marketing sources using Snowflake, dbt and Matillion. Built Airflow/Python ingestion and Redshift transformation workflows on a separate Intuita engagement.
- Comcast / Sky: developed a metadata extraction solution using Python, PostgreSQL and Collibra, and a separate tested GCP/AlloyDB prototype for metadata and database-change capture. Hyperoptic: mapped telecom data models to TM Forum SID/ODA using ER/UML modelling tools.

Further architecture, AI and education experience

Solution Architect / Technical Lead - eADR

AKD / Croatian Ministry of the Sea, Transport and Infrastructure | Nov 2025 - Aug 2026

- Designed architecture and development specifications for six modules of an EU-funded national transport platform, connecting business processes, domain models, APIs, identity/RBAC, audit trails and reporting.
- Coordinated technical delivery and stakeholder decisions; led UAT planning, integration validation and requirements-to-test traceability. Defined API contracts, error handling, retry/idempotency rules and acceptance criteria.

Data Engineering Instructor / Workshop Author

NEOS (KONČAR Group) | Apr 2026

- Designed and delivered a two-day Azure Databricks engineering workshop covering Delta Lake, Auto Loader, Structured Streaming, SQL, Unity Catalog, data quality and orchestration.
- Built banking demonstrations, guided labs and follow-up assignments, including dbt tests, Bundles and delivery workflows. Connected practical exercises with Associate/Professional certification topics and adapted material to the team's environment.

Data Architect / Technical Lead - Road Transport Analytics

AKD / Croatian Ministry of the Sea, Transport and Infrastructure | Jul - Dec 2024

- Designed an open-source lakehouse for a planned 250 TB+ scope covering 75,000+ drivers and 65,000+ vehicles. Defined 40+ safety, compliance and efficiency indicators integrating tachograph, inspection and ministry data.
- Prepared infrastructure and processing specifications, governance requirements and technical RFI/RFP inputs. Proposed Spark, MinIO, dbt, Dagster, OpenMetadata and PostgreSQL; aligned platform, engineering and analytics needs.

Data Architect / Head of AI

8tracks.com / mixer.fm | Apr - Aug 2024

- Developed AWS-based ML/ETL components for audio analysis using S3, Lambda, EFS, RDS connectivity and OpenSearch, with automated build/deployment through CodeBuild, SAM and CloudFormation. Built Python/TensorFlow audio-feature and genre-classification prototypes.

Data Engineer / Data Scientist / Lecturer

Algebra / Algebra LAB | Mar 2018 - Sep 2022

- IRI2 / Neural Grid: developed PMU ingestion, Avro schemas, Kafka/Java Streams synchronisation and rules, PostgreSQL integration and a reactive PLC interface; achieved approximately 50 ms processing-to-result latency.
- Delivered data engineering, analytics and applied ML work across research and public-sector projects; developed computer-vision demonstrations and educational material for the Erasmus ADSEE programme.

Earlier experience: Zagreb School of Economics and Management, 2002-2017 - information systems development and integration, technical leadership, digital learning and professional education.

Education and professional development

PhD in Applied Computer Science — University of Zagreb, Faculty of Electrical Engineering and Computing

Awarded 2012. Research: information-system interoperability and integration of learning platforms with digital repositories.

Udacity: Machine Learning Engineer Nanodegree; Deep Learning Nanodegree.

Previous credentials: SnowPro Core (2024; expired Jan 2026); AWS Cloud Practitioner (2022).

Languages: English - full professional proficiency, including consulting and teaching; Croatian - native.

Public Databricks workshop materials